



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Falcon Landing; D.R. Horton Atlanta West
Location: Gainesville, GA
Description: Mansfield - P, UBsmt

Date: September 22, 2024

By: TJH

House Faces: Worst West

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	3,326 ft ²
Volume	28,548 ft ³
Percentage Glass	9 %
Number of Bedrooms	5

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	11040
SEER	15
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x6	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.5	Exterior Door		R-2	
	D2	0.5	Garage Door		R-2	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling	2x4	R-38 blown	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.053	Framed		R-19 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.35	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling	Heating		
Sensible Load		33718 Btu/hr	39597 Btu/hr		
Minimum Nominal Capacity		3.75 Tons	11.60 kW		
Load Density		888 SF/Ton	3.5 W/SF		
Floor Area Served		3326.0 SF	3326.0 SF		
Mechanical Ventilation		N/A CFM	N/A CFM		
System Size		4.0 Tons	15.0 kW		
Leaving Air Temperature		57.2 deg F	99.6 deg F		
Installed Capacity		832 SF/Ton	4.5 W/SF		

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

Date: September 22, 2024
By: TJH
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[illegible]

Unit 1

Size

4.00

Tons

Airflow

1600

CFM

Heat kW

15

kW

First Floor Airflows

Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer / Hall	56	78	67	
Powder	12	13	13	
Garage Entry	34	45	39	
Pantry	15	15	15	
Bed 5 WIC	14	20	17	
Bed 5	76	112	94	
Kitchen	170	76	123	
Breakfast	47	67	57	
Family	166	172	169	
Dining	31	45	38	
Living	94	117	105	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	21	37	29	
Hall	48	42	45	
Laundry	47	9	28	
Bath 3	15	23	19	
Bed 2 & WIC	123	110	116	
Rec Rm	205	146	175	
Bed 3	98	102	100	
Bed 3 WIC	17	28	22	
Bed 4	66	70	68	
Bed 4 WIC	15	27	21	
Bed 1 WIC	26	34	30	
Bath 1 & Closet	37	40	39	
Bed 1	152	161	156	
Bath 2	14	11	12	

Project:	Falcon Landing: D.R. Horton Atlanta West	Date:	September 22, 2024	Typical Ceiling Height	9	Cooling Size	2	DX Cooling kWh	1878	
Location:	Gainesville, GA	By:	TJH	Heating Size	2	RC Heating kWh	3258			
Description:	Mansfield - P, UBsm	Level(s):	First Floor	Kneewall Height	1	Best Case	2	Total kWh	5135	
	Square Feet	1350	Volume	12150	SF/Ton	675	SEER	15		
COOLING LOAD CALCULATION										
Components				Ceiling Height		9	9	9	9	
Type	Description	CLTD (°F)	U-Value (Btu/hr-sf-F)	Cooling Load Factor	Entire House		Foyer / Hall		Powder	
					Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Glass	G1 South	Right	(E)	16 Btu/hr-sf	30	474				
	G1 SouthWest		(SE)	26 Btu/hr-sf	0	0				
	G1 West	Front	(S)	29 Btu/hr-sf	30	885				
	G1 NorthWest		(SW)	18 Btu/hr-sf	0	0				
	G1 North	Left	(W)	6 Btu/hr-sf	15	93				
	G1 NorthEast		(NW)	8 Btu/hr-sf	0	0				
	G1 East	Back	(N)	13 Btu/hr-sf	90	1179				
	G1 SouthEast		(NE)	14 Btu/hr-sf	0	0				
	G1 Sky Light			46 Btu/hr-sf	0	0				
	G1 Shaded			6 Btu/hr-sf	13	81	13	81		
Door	D1 Door	21	0.50	10.50 Btu/hr-sf	10	105	10	105		
	D2 Door	30	0.50	15.00 Btu/hr-sf	18	267				
Above Grade Walls	W1 South	16	0.08	1.23 Btu/hr-sf	362	446				
	W1 SouthWest	21	0.08	1.62 Btu/hr-sf	0	0				
	W1 West	23	0.08	1.77 Btu/hr-sf	130	229	30	54		
	W1 NorthWest	19	0.08	1.46 Btu/hr-sf	0	0				
	W1 North	13	0.08	1.00 Btu/hr-sf	166	166				
	W1 NorthEast	19	0.08	1.46 Btu/hr-sf	0	0				
	W1 East	23	0.08	1.77 Btu/hr-sf	390	690				
	W1 SouthEast	21	0.08	1.62 Btu/hr-sf	0	0				
	W1 South	16	0.08	1.23 Btu/hr-sf	0	0				
	W1 SouthWest	21	0.08	1.62 Btu/hr-sf	0	0				
Basement Walls	W1 West	23	0.08	1.77 Btu/hr-sf	0	0				
	W1 NorthWest	19	0.08	1.46 Btu/hr-sf	0	0				
	W1 North	13	0.08	1.00 Btu/hr-sf	0	0				
	W1 NorthEast	19	0.08	1.46 Btu/hr-sf	0	0				
	W1 East	23	0.08	1.77 Btu/hr-sf	0	0				
	W1 SouthEast	21	0.08	1.62 Btu/hr-sf	0	0				
	W1 South	16	0.08	1.23 Btu/hr-sf	0	0				
	W1 SouthWest	21	0.08	1.62 Btu/hr-sf	0	0				
	W1 West	23	0.08	1.77 Btu/hr-sf	0	0				
	W1 NorthWest	19	0.08	1.46 Btu/hr-sf	0	0				
Other	W1 North	13	0.08	1.00 Btu/hr-sf	0	0				
	W1 NorthEast	19	0.08	1.46 Btu/hr-sf	0	0				
	W1 East	23	0.08	1.77 Btu/hr-sf	0	0				
	W1 SouthEast	21	0.08	1.62 Btu/hr-sf	0	0				
	W2 Grade	0	0.07	0.00 Btu/hr-sf	0	0				
Roof	W1 Partition	10	0.08	0.77 Btu/hr-sf	0	0				
	W1 Garage	30	0.08	2.31 Btu/hr-sf	529	1222	192	443	57	131
	W3 Attic	55	0.05	2.92 Btu/hr-sf	0	0				
	W3 Kneewall	55	0.05	2.92 Btu/hr-sf	0	0				
	R1 Ceiling	55	0.03	1.43 Btu/hr-sf	0	0				
Floor	R2 Vaulted	55	0.05	2.92 Btu/hr-sf	0	0				
	F1 Crawl/Uncon.	11	0.05	0.58 Btu/hr-sf	1350	787	128	75	34	20
	F1 Garage	30	0.05	1.59 Btu/hr-sf	0	0				
	F2 Slab	0	0.10	0.00 Btu/hr-sf	0	0				
	Internal People			255 Btu/hr	2	510				
Vent.	Internal Lights (W)			3.413 Btu/hr-W	678	2314	64	218	17	58
	Internal Appliances (W)			3.413 Btu/hr-W	930	3174				
	Infiltration Volume	0.25	ACH	0.09 Btu/hr-cf		1094		104		28
	Area			CF	12150		1152		306	540
				SF	1350		128		34	60
Subtotal (Btu/hr)						13716		1079		236
Safety Factor						1372				
Sensible Cooling Load (Btu/hr)						15087				
Nominal Tons Required (75% sens. cap.)						1.68				
System Selection					2 Tons	Space CFM	800	63	14	38
HEATING LOAD CALCULATION										
Components				Heating Load Factor		Entire House		Foyer / Hall		
Type	Description	ΔT _f	U _f	V _f	A _f	Qty	Btu/hr	Qty	Btu/hr	
						Qty	Btu/hr	Qty	Btu/hr	
Glass	G1	54	0.35	18.9	Btu/hr-sf	178	3371	13	246	
	G2	54	0.00	0.0	Btu/hr-sf	0	0	0	0	
	G3	54	0.00	0.0	Btu/hr-sf	0	0	0	0	
Door	D1	54	0.50	27.0	Btu/hr-sf	10	270	10	270	
	D2	54	0.50	27.0	Btu/hr-sf	18	481	0	0	
Walls	W1	54	0.08	4.2	Btu/hr-sf	1047	4352	30.3	126	
	W2	54	0.07	3.6	Btu/hr-sf	0	0	0	0	
	W3	54	0.05	2.9	Btu/hr-sf	0	0	0	0	
	W1 Partition	27	0.08	2.1	Btu/hr-sf	529	1100	192	399	
	W2 Grade	27	0.07	1.8	Btu/hr-sf	0	0	0	0	
	R1	54	0.03	1.4	Btu/hr-sf	0	0	0	0	
	R2	54	0.05	2.9	Btu/hr-sf	0	0	0	0	
	F1	54	0.05	2.9	Btu/hr-sf	1350	3864	128	366	
	F2 Slab	27	0.10	2.7	Btu/hr-sf	0	0	0	0	
	Area				SF	1350		128	34	
Floor	Infiltration	0.25	ACH	0.24	Btu/hr	0	2952		280	
	Subtotal (Btu/hr)						16390		1687	
	Safety Factor						2458			
	Heat Loss (Btu/hr)						18848			
	Minimum Heater Size (kW)						5.52			
System Selection					8 kW	Htg. CFM	800	82	14	

Project: Falcon Landing; D.R. Horton Atlanta West

Location: Gainesville, GA

Description:	<u>Mansfield - P, UBsmt</u>
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Date: September 22, 2024

By: TJH

Level(s): Second Floor

Typical Ceiling Height	8
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Cooling Size 2

DX Cooling kWh	2318
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Heating Size 2

RC Heating kWh	3586
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Best Case	2
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Total kWh	5905
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Square Feet	1976	Volume	16398	SF/Ton	988
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SEER 15

COOLING LOAD CALCULATION										Ceiling Height		18		8		8		8		8		8		8		8		8		8		8		8			
Components					CLTD (°F)	U-Value (Btu/hr-sq-ft-F)	Cooling Load Factor	Entire House		Stairs		Hall		Laundry		Bath 3		Bed 2 & WIC		Rec Rm		Bed 3		Bed 3 WIC		Bed 4		Bed 4 WIC		Bed 1 WIC		Bath 1 & Closet		Bed 1		Bath 2	
	Type	Description		Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G1	South	Right	(E)	16	Btu/hr-sf	30	474																													
	G1	SouthWest		(SE)	26	Btu/hr-sf	0	0																													
	G1	West	Front	(S)	29	Btu/hr-sf	75	2212							30	885	30	885	15	442																	
	G1	NorthWest		(SW)	18	Btu/hr-sf	0	0																													
	G1	North	Left	(W)	6	Btu/hr-sf	15	93																													
	G1	NorthEast		(NW)	8	Btu/hr-sf	0	0																													
	G1	East	Back	(N)	13	Btu/hr-sf	30	392																													
	G1	SouthEast		(NE)	14	Btu/hr-sf	0	0																													
	G1	Sky Light			46	Btu/hr-sf	0	0																													
G1	Shaded			6	Btu/hr-sf	0	0																														
Door	D1	Door	21	0.50	10.50	Btu/hr-sf	0	0																													
	D2	Door	55	0.50	27.50	Btu/hr-sf	0	0																													
Above Grade Walls	W1	South	16	0.08	1.23	Btu/hr-sf	414	510	73	90	-	16	0.08	-	75	92	116	143	18	22	-	-	-	-	-	-	-	-	-	-	-	-	132	163	-	-	
	W1	SouthWest	21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West	23	0.08	1.77	Btu/hr-sf	357	632	-	-	-	-	-	-	-	-	132	234	122	217	103	182	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthWest	19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	North	13	0.08	1.00	Btu/hr-sf	390	390	-	-	-	-	-	-	-	-	9	9	18	18	113	113	76	76	115	115	60	60	-	-	-	-	-	-			
	W1	NorthEast	19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East	23	0.08	1.77	Btu/hr-sf	402	712	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast	21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South	16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Basement Walls	W1	SouthWest	21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West	23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthWest	19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	North	13	0.08	1.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthEast	19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East	23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast	21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South	16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest	21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Other	W2	Grade	0	0.07	0.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	Partition	10	0.08	0.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	Garage	30	0.08	2.31	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W3	Attic	55	0.05	2.92	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W3	Kneewall	55	0.05	2.92	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Floor / Roof	R1	Ceiling	55	0.03	1.43	Btu/hr-sf	1976	2826	59	84	207	296	58	83	52	74	216	309	308	440	164	235	46	66	189	270	36	51	78	112	140	200	353	505	70	100	
	R2	Vaulted	55	0.05	2.92	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Crawl/Uncon.	11	0.05	0.58	Btu/hr-sf	52	30	52	30	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Garage	30	0.05	1.59	Btu/hr-sf	629	1000	-	-	72	114	-	-	-	-	-	308	490	164	261	46	73	39	62	-	-	-	-	-	-	-	-	-			
	F2	Slab	0	0.10	0.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Internal	People				255	Btu/hr	5	1275	-	-	-	-	-	-	1	255	1	255	1	255	1	255	-	1	255	-	-	-	-	-	-	1	255	-	-		
	Lights (W)				3.413	Btu/hr-W	990	3379	30	102	104	355	29	99	26	89	108	369	154	526	82	280	23	78	95	324	18	61	39	133	70	239	177	604	35	119	
	Appliances (W)				3.413	Btu/hr-W	450	1536	-	-	-	200	683	-	-	-	-	250	853	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	Infiltration		0.25	ACH	0.09	Btu/hr-cf		1476	-	96	149	-	42	-	37	-	156	-	222	-	118	-	33	-	136	-	26	-	56	-	101	-	254	-	50		
Vent	Volume				CF		16398	1062		1656		464		416		1728	2464		1312		368		1512		288		624		1120		2824		560				
	Area				SF		1976	59		207		58		52		216	308		164		46		189		36		78		140		353		70				
Subtotal (Btu/hr)								16937		402		914		906		293		2359		3927		1885		326		1256		285		497		710		2905		270	
Safety Factor								1694																													
Sensible Cooling Load (Btu/hr)								18631																													
Nominal Tons Required (75% sens. cap.)								2.07																													
System Selection 2 Tons Space CFM								800		19		43		43		14		111		186		89		15		59		13		23		34		137		13	
HEATING LOAD CALCULATION										Ceiling Height		18		8		8		8		8		8		8		8		8		8		8		8			
Components					ΔT (°F)	U-Value (Btu/hr-sq-ft-F)	Heating Load Factor	Entire House		Stairs		Hall		Laundry		Bath 3		Bed 2 & WIC		Rec Rm		Bed 3		Bed 3 WIC		Bed 4		Bed 4 WIC		Bed 1 WIC		Bath 1 & Closet		Bed 1		Bath 2	
	Type	Description		Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G2			54	0.35	18.9	Btu/hr-sf	150	2835	0	-	0	-	0	-	0	-	30	567	30	567	15	284	0	-	15	284	0	-	0	-	0	-	60	1,134	0	-
	G2			54	0.00	0.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	G3			54	0.00	0.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
Door	D1			54	0.50	27.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	D2			54	0.50	27.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
Walls	W1			54	0.08	4.2	Btu/hr-sf	1563	6499	72.8	303	0	-	0	-	75	312	257	1,070	158	658	215	895	75.8	315	115	477	109	452	111	461	95.9					